1.

Solving Differential Equation:

$$E * I * \frac{d^4 u_y}{dx^4} = q_0$$

$$E * I * \frac{d^3 u_y}{dx^3} = q_0 * x + C_1$$

$$E * I * \frac{d^2 u_y}{dx^2} = q_0 * \frac{x^2}{2} + C_1 * x + C_2$$

$$E * I * \frac{du_y}{dx} = q_0 * \frac{x^3}{6} + C_1 * \frac{x^2}{2} + C_2 * x + C_3$$

$$E * I * u_y = q_0 * \frac{x^4}{24} + C_1 * \frac{x^3}{6} + C_2 * \frac{x^2}{2} + C_3 * x + C_4$$

1. $At\ x = L, \frac{d^3 u_y}{dx^3} = 0;\ Thus: P = q_0 * L + C_1 \rightarrow C_1 = P - q_0 * L$
2. $At\ x = L, \frac{d^2 u_y}{dx^2} = 0;\ Thus: 0 = q_0 * \frac{L^2}{2} + C_1 * L + C_2 \rightarrow$
   $0 = q_0 * \frac{L^2}{2} + (P - q_0 * L\ ) * L + C_2 \rightarrow C_2 = \frac{q_0 L^2}{2} - L * P$
3. $At\ x = 0, \frac{du_y}{dx} = 0;\ Thus: 0 = q_0 * \frac{0^3}{6} + C_1 * \frac{0^2}{2} + C_2 * 0 + C_3 \rightarrow C_3 = 0$
4. $At\ x = 0, u_y = 0;\ Thus: 0 = q_0 * \frac{0^4}{24} + C_1 * \frac{0^3}{6} + C_2 * \frac{0^2}{2} + C_3 * 0 + C_4 \rightarrow$
   $C_4 = 0$

$$\boldsymbol{E * I * u_y = q_0 * \frac{x^4}{24} + (P - q_0 * L) * \frac{x^3}{6} + \left(\frac{q_0 L^2}{2} - L * P\right) * \frac{x^2}{2}}$$